

Lecture notes on QFT

Giacomo Galliani, course held by prof. Raul Ronscht

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1

Lorentz and Poincarè symmetries

1.1 The Lorentz group

Quantum field theory is the theory that emerges when one wishes to combine special relativity with quantum mechanics. As such, it is important to study the Lorentz and Poincarè symmetries, within their appropriate mathematical framework, that is, group theory.

The Lorentz group is defined as the group of *linear transformations*, applied on a 4-vector in Minkowski spacetime, that we usually indicate with x^μ , that leaves invariant its "square" $x^\mu x_\mu = g_{\mu\nu} x^\mu x^\nu = (x^0)^2 - (\vec{x})^2$. (here, $g_{\mu\nu} = \text{diag}(+, -, -, -)$). The condition of linearity means that we can choose a linear representation, in terms of which we can explicitly write the transformation as

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu_\nu x^\nu \quad (1.1)$$

In general, mathematicians call the linear group of transformation that leave invariant $\sum^m x_i^2 - \sum^n y_i^2$ $O(n, m)$. Thus, the Lorentz group is $O(3, 1)$. Now the fundamental condition of the invariant square means that

$$x'^\mu x'_\mu = g_{\mu\nu} (\Lambda^\mu_\rho x^\rho) (\Lambda^\nu_\sigma x^\sigma) = g_{\rho\sigma} x^\rho x^\sigma \quad (1.2)$$

that is,

$$g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = g_{\rho\sigma} \quad (1.3)$$

Every element of a group must have an inverse, such that multiplication by left or right gives back the identity. What is the inverse of a Lorentz transformation? Take again the condition of leaving the square invariant:

$$g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = \Lambda_{\nu\rho} \Lambda^\nu_\sigma = g_{\rho\sigma} \implies \Lambda_{\nu}^{\rho'} \Lambda^\nu_\sigma = g_{\rho\sigma} g^{\rho\rho'} = \delta_\sigma^{\rho'} \quad (1.4)$$

by definition, then, the inverse is given by

$$(\Lambda^{-1})^\rho_\nu = \Lambda^\rho_\nu \quad (1.5)$$

In matrix form, we would write the condition of the square invariant as $g = \Lambda^T g \Lambda$. Da questa, è evidente che

$$\det(\Lambda^2) = 1 \implies \det(\Lambda) = \pm 1 \quad (1.6)$$

Now the set of transformations that have determinant one forms a subgroup of $O(3, 1)$ called the proper transformation, which is denoted as $SO(3, 1)$. In particular, we have that infinitesimal transformations, that is transformation close to identity, are always *proper* transformations.

There is also another distinction. It is easy to check that Λ_0^0 must obey $\Lambda^2 \geq 1$. That is, we can divide transformations where $\Lambda_0^0 > 1$, and $\Lambda_0^0 < -1$. The first ones are called *orthocronus*. From now on, when we will refer to Lorentz group as the proper and orthocronus subgroup of $O(3, 1)$.

Now consider an infinitesimal transformation, that we will write as

$$\Lambda_\nu^\mu = \delta_\nu^\mu + \omega_\nu^\mu \quad (1.7)$$

this one also has to leave the square invariant, so it holds

$$g_{\rho\sigma} = g_{\mu\nu}(\delta_\rho^\mu + \omega_\rho^\mu)(\delta_\sigma^\nu + \omega_\sigma^\nu) \quad (1.8)$$

$$= g_{\rho\sigma} + g_{\mu\sigma}\omega_\rho^\mu + g_{\rho\nu}\omega_\sigma^\nu + o(\omega^2) \quad (1.9)$$

$$= g_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} \quad (1.10)$$

Thus, $\omega_{\mu\nu}$ is antisymmetric. Since it is a 4×4 matrix, this means that it has 6 free parameters. This is right, as the six parameters correspond to the 3 rotations and the 3 boosts that make up the Lorentz group. Introducing a parameter η , called *rapidity*, one can think as boost as *hyperbolic* rotations, in the sense that they can be written as

$$t \rightarrow t \cosh \eta + x \sinh \eta \quad (1.11)$$

$$x \rightarrow t \sinh \eta + x \cosh \eta \quad (1.12)$$

Thus, each element of the group is written in a differentiable function of the six parameters. Mathematicians call this type of group a Lie group.

Now recall that in quantum mechanics, symmetries were realized by some operators acting on a state. These operators are a *representation* of the group. A *linear representation* of the group, is a map that assigns each element of the group a matrix, $g \in G \rightarrow D(g)$, that preserve the group structure, that is, $D(g_1 g_2) = D(g_1) D(g_2)$, $\forall g_1, g_2 \in G$.

Lie groups are special, in the sense that they have a special representation: we can always write

$$D(g(\theta)) = e^{i\theta^a T^a} \quad (1.13)$$

where the elements $\{T^a\}$ are called *generators* of the group. Since this representation must preserve the group structure, one finds that generators must obey some *commutation relations*:

$$[T^a, T^b] = i f_c^{ab} T^c \quad (1.14)$$

where, surprisingly, the structure constants f do not depend on the particular representation. The generators then live in a vector space, equipped with a bilinear closed form (that is, a bilinear map $V \times V \rightarrow V$). This place is called a *Lie algebra*. For the Lorentz group, we will then write a general transformation as

$$\Lambda = \exp \left(-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu} \right) \quad (1.15)$$

where the antisymmetric matrix $\omega_{\mu\nu}$ carries the six parameters, (the "angles" of rotation and boosts), and $J^{\mu\nu}$ are the generators. To avoid causing confusion, we stress that the generators are elements of a vector space, so that each element of $J^{\mu\nu}$ lives in a vector space. Don't be surprised then when we will use this object for some computations, and we will use more indexes on it, such as $(J^{\mu\nu})_{\sigma}^{\rho}$. Of course, if this was not the case than Λ would be a scalar, with as little physical sense.¹

When we select a representation, if we have an object, say a field, ϕ^i , this transforms under the action of the group as

$$\phi^i \rightarrow \phi'^i = \left(e^{-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}} \right)_j^i \phi^j \quad (1.16)$$

One can then *classify* the objects, or the fields, based on how they transform under the action of the group.

Before we start, there is a useful way to treat the generators of the Lorentz group. If one writes

$$J^i = \frac{1}{2} \epsilon^{ijk} J^{jk}, \quad K^i = J^{0i} \quad (1.17)$$

one then finds the following commutation relations:

$$[J^i, J^j] = i\epsilon^{ijk} J^k, \quad [J^i, K^j] = i\epsilon^{ijk} K^k, \quad [K^i, K^j] = i\epsilon^{ijk} J^k \quad (1.18)$$

The first commutation relation is the usual angular momentum commutation relation from quantum mechanics. In Lie algebras, this is called a $\mathfrak{su}(2)$ sub-algebras, which is due to the fact that rotations are a subgroup of the Lorentz group.

The second commutation relation, shows that K^i behaves as a 3-vector under rotations. Finally, the last commutation tells that the combination of two boosts is equivalent to a rotation.

Since we now understand that J and K perform a very different role in the Lorentz group, it is useful to write a general Lorentz transformation in a way that is more transparent: define

$$\theta^i = -\frac{1}{2} \epsilon^{ijk} \omega^{jk}, \quad \eta^i = \omega^{i0} \quad (1.19)$$

then, θ^i are the angles of the rotations, and η^i are the 3 rapidities, associated with the 3 boosts. A general Lorentz transformation is then given by

$$\Lambda = e^{(-i\vec{\theta} \cdot \vec{J} + i\vec{\eta} \cdot \vec{K})} \quad (1.20)$$

¹Strictly speaking, it has a mathematical sense. Every group has a very simple representation, called the trivial representation, where every group element is mapped into the identity. The representation is a valid one, because 1 times 1 is always 1 and then the group structure is preserved. This may seem a useless fact, but I think it's interesting, even for physicists, because it shows how not every representation of the group is a useful one.

Where with the dot-product $\cdot$ we mean the usual euclidean inner product in $\mathbb{R}^3$.

Now let's look at how tensors transform. Tensor, by definition, are object that transform as

$$T^{\mu\nu} \rightarrow T'^{\mu\nu} = \Lambda_{\sigma}^{\mu} \Lambda_{\rho}^{\nu} T^{\sigma\rho} \quad (1.21)$$

this is a first example of a *reducible* representation. What this means is that a tensor is made up of pieces, that transform independently under Lorentz transformations.

To see why this is the case for tensors, let's split the tensor in symmetric and antisymmetric:

$$T^{\mu\nu} = S^{\mu\nu} + A^{\mu\nu}, \quad S^{\mu\nu} = \frac{1}{2}(T^{\mu\nu} + T^{\nu\mu}), \quad A^{\mu\nu} = \frac{1}{2}(T^{\mu\nu} - T^{\nu\mu}) \quad (1.22)$$

Now, $S^{\mu\nu}$ is the sum of its trace and a symmetric traceless tensor. The trace is given by

$$S = g_{\mu\nu} S^{\mu\nu} \quad (1.23)$$

since it is a contraction of all indices, it is clearly a Lorentz scalar. It is a 1 dimensional representation of the group. The anti-symmetric part is instead a 6-dimensional representation, and the symmetric traceless is 9 dimensional. In the formalism of group theory, one writes

$$4 \times 4 = 6 + 1 + 9 \quad (1.24)$$

Not only we can check how some objects transform under the action of $SO(3,1)$, but also it becomes interesting to classify the fields, based on how they behave under the very important subgroup $SO(3)$, the group of rotations.

Representations of $SO(3)$ are labeled by two integers, j, j_z , with $j_z = -j, \dots, j$. A scalar has $j = 0$. Instead a 4-vector is a reducible representation of $SO(3)$. This is because rotation acts only on the spacial components of V^{μ} . Then we will write

$$4 = 1 + 3 \quad (1.25)$$

where the 1 dimensional representation has $j = 0$, the 3-dimensional has $j = 1$. (in general, the dimension of the representation is given by $2j + 1$).

What of a tensor? If we think of a tensor as a tensor product of two vectors, then it is natural to write

$$4 \times 4 = (1 + 3) \times (1 + 3) = 1 \times 1 + 1 \times 3 + 3 \times 1 + 3 \times 3 \quad (1.26)$$

This result has at least two interesting interpretations:

1. In terms of the adding rules of angular momenta.

We know that a 4-vector carries a $j = 0$ and $j = 1$ representation. Then, when we add two of those up, we get:

- (a) A singlet state, given by $j_1 = 0, j_2 = 0$. This gives the dimension 1 representation.
- (b) Two triplet states, given by $j_1 = 1, j_2 = 0$ and $j_1 = 0, j_2 = 1$.

- (c) The last collection of $j_1 = 1, j_2$ states, that is the tensor product of 3-dimensional representation, i.e. a 9-dimensional representation.
2. In terms of how rotation acts on different parts of the tensor. If we take T^{00} , then it is a scalar, so we find the first 1-dimensional rep. Then, T^{i0} and T^{0i} are going to be vectors under rotations, and that explains the two 3-dimensional rep. Finally, T^{ij} is a 3-tensor, so a tensor product of two 3-vectors, and that explains the 9-dimensional rep.

1.2 Spinorial representation

QFT is about particles, and particles in general carry a spin. We know since quantum mechanics that spin is better described by $SU(2)$ rather than $SO(3)$. The difference between the two groups is subtle. Their algebra is the same, and that reflects the fact that the two groups are *locally* isomorphic. However, under a 2π rotation, an electron (or a spin 1/2 particle) carries a minus sign in his wave function. Since mathematicians know that $SO(3) = SU(2)/\mathbb{Z}_2$, we will use $SU(2)$.

Another consequence of this is that rep. of $SU(2)$ are labeled by *half-integer* labels, $j = 0, 1/2, 1, \dots$. The representation $j = 1/2$ is called *fundamental*, because it turns out that all representation can be constructed by $j = 1/2$ pieces. For example if i combine two $j = 1/2$ spin, is well known that i get

$$1/2 \times 1/2 = 0 + 1 \quad (1.27)$$

$SU(2)$ representation become even more important in qft because of a special feature of the Lorentz algebra: if we define

$$J^{\pm, i} := \frac{J^i \pm iK^i}{2} \quad (1.28)$$

then the relation commutations become

$$[J^{+i}, J^{+j}] = i\epsilon^{ijk}J^{+k}, [J^{-i}, J^{-j}] = i\epsilon^{ijk}J^{-k}, [J^{+i}, J^{-j}] = 0 \quad (1.29)$$

Basically, we have shown that we can divide the Lorentz algebra in two $\mathfrak{su}(2)$ subalgebras. This means that we can label representations by a pair of two half integers j_1, j_2 .

Let's now study their most important properties. We start with the easiest one, the scalar representation of dimension 1, witch has $j_1 = 0, j_2 = 0$. This has $J^i = 0, K^i = 0$. Then we have two different representation of dimension 2, that we will label as $(\frac{1}{2}, 0), (0, \frac{1}{2})$. Those are the left handed and right handed Weyl spinors, denoted typically as ψ_L, ψ_R . In those, $J^+ = 0$ and J^- can be rapresented by 2×2 complex matrixes (the Pauli matrixes), or the other way around. Now that we know this, we can invert the relations for $J^\pm$ and write rotation and boosts in terms of Pauli matrixes:

$$J^i = J^{+i} + J^{-i} = \frac{\sigma^i}{2} \quad (1.30)$$

$$K^i = -i(J^+ - J^-) = i\frac{\sigma^i}{2} \quad (1.31)$$

Thus we find the rules of transformation for Weyl spinors:

$$\psi_L \rightarrow \psi'_L = \Lambda_L \psi_L, \Lambda_L = \exp\left(i(-\vec{\theta} \cdot \vec{J} + \vec{\eta} \cdot \vec{K})\right) = \exp\left((-i\vec{\theta} - \vec{\eta}) \cdot \frac{\vec{\sigma}}{2}\right) \quad (1.32)$$

Similarly, one can find the rule of transformation for right handed spinors, given by

$$\Lambda_R = \exp \left((-i\vec{\theta} + \vec{\eta}) \cdot \frac{\vec{\sigma}}{2} \right) \quad (1.33)$$

Finally, the $(\frac{1}{2}, \frac{1}{2})$ representation is a 4-dimensional complex representation, that can be thought has the composition of a Weyl left handed spinor and a Weyl right handed spinor.

1.3 Field representation

In general, how does a *field* transform under lorentz transformations? We can classify field depending on how they transform.

The simplest case is the scalar field. Since it is a scalar, $\phi'(x') = \phi(x)$. However, in general $\phi'(x) \neq \phi(x)$. Let us then compute the difference. The coordinate shift, if we take an infinitesimal transformation is given by

$$x^\mu \rightarrow x'^\mu = x^\mu + \delta x^\mu, \quad \delta x^\mu = -\frac{i}{2} \omega_{\sigma\nu} (J^{\sigma\nu})^\mu \quad (1.34)$$

Then we find that

$$\delta_0 \phi := \phi'(x) - \phi(x) \simeq -\delta x^\mu (\partial_\mu \phi) \quad (1.35)$$

Now, if we try and put this in the form

$$\delta_0 \phi = -\frac{i}{2} \omega_{\mu\nu} L^{\mu\nu} \phi \quad (1.36)$$

we find that the $L^{\mu\nu}$ operators are

$$L^{\mu\nu} = i(x^\nu \partial^\mu - x^\mu \partial^\nu) = x^\mu p^\nu - x^\nu p^\mu \quad (1.37)$$

if like in QM we call $p^\mu = i\partial^\mu$. This generators obey the lorentz algebra, and form a representation of the algebra, using the field as a basis.

Note that clearly if we take only the spacial components, than the operator is nothing but the angular momentum operator for quantum mechanics.

Let us than do the same type of computation for a Weyl spinor. We can already guess what will be the outcome: the total angular momentum that we will find will be the sum of an orbital angular momentum, and a spin. Lets see: the only difference now is that ψ_L is not a scalar anymore, but rather transforms with the relation derived in the previous section. We can than compute

$$\delta_0 \psi_L = \psi'_L(x) - \psi_L(x) \quad (1.38)$$

$$\simeq \psi'_L(x') - \psi_L(x) - \delta x^\rho \partial_\rho \psi_L(x) \quad (1.39)$$

$$\simeq -\frac{i}{2} \omega_{\mu\nu} S^{\mu\nu} - \delta x^\rho \partial_\rho \psi_L \quad (1.40)$$

Again, if we try and write this as

$$\delta_0 \psi_L = -\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu} \psi_L \quad (1.41)$$

we find, as promised,

$$J^{\mu\nu} = S^{\mu\nu} + L^{\mu\nu} \quad (1.42)$$

because if we take the spacial components and define

$$S^i = \frac{1}{2} \epsilon^{ijk} S^{jk} \quad (1.43)$$

this is the old spin operator from QM.

Dirac fields are then fields ψ in the $(1/2, 1/2)$ representation. At first, it might not be obvious why we need them. One main reason is that a right-handed spinor is mapped in a left-handed spinor by parity. This is because K^i is a true vector under rotations, and so $K^i \xrightarrow{P} K^i$, while J^i is a pseudovector, and is mapped in itself by parity.

If we want to make a theory that admits parity as a symmetry (e.g. electromagnetism), then we'll have to take a Dirac field that under parity is then simply transformed as

$$\psi(x) \rightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \psi(x) \quad (1.44)$$

1.4 Poincarè group

We now want to enlarge the Lorentz group, by adding spacetime translations, of the form

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu \quad (1.45)$$

the resulting group is called the Poincarè group.

Given p^μ the generators of translations, we can find the commutation relations with the other generators and thus write down the whole Poincarè algebra. The main results of this computation are the following:

1. J^i, P^i, K^i all behave like vectors under spacial rotations.
2. J^i, P^i commute with P^0 , and thus are conserved. This is not the case however for K^i , that does not commute with P^0 .

Let's now give a representation of P^μ , on a basis of fields. This is done in the usual way:

$$\phi'(x' - \epsilon) = e^{-ip^\mu(-\epsilon_\mu)} \phi'(x) \simeq \phi'(x) + ip^\mu \epsilon_\mu \phi'(x) \quad (1.46)$$

but also

$$\delta_o \phi = \phi'(x) - \phi(x) \simeq -\epsilon^\mu \partial_\mu \phi \quad (1.47)$$

comparing the two expressions, we get

$$p^\mu = i\partial^\mu \quad (1.48)$$

just as in QM.

2

The scalar field

2.1 The classical scalar field and the KG equation

Let's consider a field ϕ , which is a scalar under Poincaré symmetries. To have a theory from it, we have to build a lagrangian, specifically a lorentz invariant lagrangian. Then we can write the equation of motion, and solve it to derive the properties of the field.

To be able to write second order e.o.m., we need to have a "kinetic" term in the lagrangian, which is quadratic in the field first derivative. Since we need it to be a lorentz scalar, we'll write $\partial_\mu \phi \partial^\mu \phi$. Then we add a potential term. The easiest way to do it is by adding a quadratic term in the field, so the whole lagrangian reads

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) \quad (2.1)$$

the parameter m is interpreted as the mass of the field, for reasons that will become apparent in a moment. If one computes the euler-lagrange equation of motion for this lagrangian, one finds the Klein-Gordon equation:

$$(\partial_\mu \partial^\mu + m^2)\phi = (\square + m^2)\phi = 0 \quad (2.2)$$

This is just a wave equation, and the simplest solution is given by plane waves $e^{\pm i p^\mu x_\mu}$, if and only if $p^\mu p_\mu = m^2$ ¹, i.e. they satisfy the mass-shell relation. This is the reason why we think of m as the mass of the field.

To get the most general solution, one integrates all possible plane waves:

$$\phi(x) = \int \frac{d^3 p}{(2\pi)^3 \sqrt{2E_p}} (a_p e^{-ipx} + a_p^* e^{ipx}) \quad (2.3)$$

at this level, a_p is just a complex coefficient. We will see that this coefficient plays a key role when we will quantize this field. Also, note that in this expression, lorentz invariance is not self evident. However, we will show later that the measure $\frac{d^3 p}{\sqrt{2E_p}}$ is a lorentz scalar.

Observe that one can obtain the hamiltonian,

$$\mathcal{H} = \frac{1}{2}(\pi^2 + (\nabla \phi)^2 + m^2 \phi^2) \quad (2.4)$$

¹of course, to see why, just plug the candidate solution in the equation. Then one finds that $\partial_\mu \phi = \pm i p_\mu \phi$, and then $\square \phi = -p_\mu p^\mu \phi$.

and also the energy-momentum tensor:

$$T^{\mu\nu} = (\partial^\mu \phi)(\partial^\nu \phi) - g^{\mu\nu} \mathcal{L} \quad (2.5)$$

We can also compute the conserved currents, due to the lorentz symmetries. After some algebra, one finds that

$$J^{\rho\sigma\mu} = x^\rho T^{\sigma\mu} - x^\sigma T^{\mu\rho} \quad (2.6)$$

to see some of the conserved charges, take $\rho = i$, $\sigma = j$, and let's compute the noether charge:

$$M^{ij} = \int d^3x J^{ij0} = i \int d^3x \partial_0 \phi L_{ij} \phi \quad (2.7)$$

if $L_{ij} = i(x^i \partial^j - x^j \partial^i)$, the generators of rotations in the field representation. If we introduce the scalar product of two fields, as

$$\langle \phi_1 | \phi_2 \rangle = \frac{i}{2} \int d^3x \phi_1 (\partial_0 \phi_2) - (\partial_0 \phi_1) \phi_2 := \frac{i}{2} \int d^3x \phi_1 \overleftrightarrow{\partial}_0 \phi_2 \quad (2.8)$$

we find that the conserved charge is the expected value of the generators of the rotations:

$$M^{ij} = \langle \phi | L^{ij} | \phi \rangle \quad (2.9)$$

2.2 Canonical quantisation of the real scalar field

Recall that when one wishes to go from classical mechanics to quantum mechanics, at least in the canonical quantisation, he has to take the position and momentum variables q_i, p_i and promote them to operators, that obey the famous commutation relations:

$$[q_i, p_j] = i\delta^{ij} \quad (2.10)$$

where we have set $\hbar = 1$.

If one wishes then to quantize fields in the canonical way, he has to take the field $\phi(t, \vec{x})$ and its momentum $Pi(t, \vec{x})$, promote them to operators, and require that they must obey

$$[\phi(t, \vec{x}), \Pi(t, \vec{x}')] = i\delta(\vec{x} - \vec{x}') \quad (2.11)$$

Let us then start with a real scalar field, with lagrangian

$$\mathcal{L}_{KG} = \frac{1}{2}((\partial_\mu \phi)(\partial^\mu \phi) - m^2 \phi^2) \implies (\square + m^2)\phi = 0 \quad (2.12)$$

We know that the most general solution of the equation of motion can be written as

$$\phi(t, \vec{x}) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_p}} (a_{\vec{p}} e^{-ipx} + a_{\vec{p}}^\dagger e^{ipx}), \quad E_p = \sqrt{\vec{p}^2 + m^2} \quad (2.13)$$

Since now the field has become an operators, also $a, a^\dagger$ are operators. Imposing the commutation relation for the fields and it's time derivative, we find the commutation relation also for them:

$$[a_{\vec{p}}, a_{\vec{q}}^\dagger] = (2\pi)^3 \delta(\vec{p} - \vec{q}), \quad [a_{\vec{p}}, a_{\vec{q}}] = 0, \quad [a_{\vec{p}}^\dagger, a_{\vec{q}}^\dagger] = 0 \quad (2.14)$$

so those are the ladder operators that we know and love from quantum mechanics.

From this, we create our *Fock space*. First we define the vacuum state, as the normalized state that gets annihilated by the destruction operators:

$$a_{\vec{p}} |0\rangle = 0, \quad \forall \vec{p}, \quad \langle 0 | 0 \rangle = 1 \quad (2.15)$$

By acting on the vacuum with construction operators, we get particle states. The general rule is

$$|\vec{p}_1, \dots, \vec{p}_n\rangle = \sqrt{2E_{\vec{p}_1}} \dots \sqrt{2E_{\vec{p}_n}} a_{\vec{p}_1}^\dagger \dots a_{\vec{p}_n}^\dagger |0\rangle \quad (2.16)$$

The terms $\sqrt{2E_{p_i}}$ are just normalization constants.

One nice thing about canonical quantisation is that is pretty straightforward: just take what you know about quantum mechanics, and now apply it to fields. The not so nice thing about canonical quantisation is that it doesn't seem to treat time and space on an equal footing. For example, the expression of the field doesn't *feel* lorentz-invariant. Let's check if it actually is:

Start with the surely covariant expression

$$\frac{d^4k}{(2\pi)^4} \delta(k^2 - m^2) (2\pi) \theta(k_0) \quad (2.17)$$

Now let's split the time and spatial components:

$$\frac{d^3k}{(2\pi)^3} dk_0 \delta(k_0^2 - (\vec{k}^2 + m^2)) \theta(k_0) = \frac{d^3k}{(2\pi)^3} dk_0 \frac{\delta(k_0 - \sqrt{k^2 + m^2})}{2k_0} \quad (2.18)$$

where we have used the known properties of the delta function. Now if one integrates the expression with respect to time gets

$$\frac{d^3k}{(2\pi)^3 2E_k} \quad (2.19)$$

we conclude that this measure is lorentz invariant. Notice that this is not quite what we have on the field: however this is not important, because when we act on the field on a state, say the vacuum, we get

$$\phi |0\rangle = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_p}} e^{ipx} a^\dagger |0\rangle \quad (2.20)$$

$$= \int \frac{d^3p}{(2\pi)^3 (2E_p)} e^{ipx} |p\rangle \quad (2.21)$$

and this expression *is* Lorentz invariant.

We could compute the hamiltonian of the theory:

$$\mathcal{H} = \frac{1}{2}(\pi^2 + (\vec{\nabla}\phi)^2 + m^2\phi^2) \quad (2.22)$$

And if we put the explicit formula for the field in terms of the ladder operators, we find (after some algebra)

$$\mathcal{H} = \int \frac{d^3p}{(2\pi)^3} \left(\frac{E_p}{2} \right) (a_p^\dagger a_p + a_p a_p^\dagger) \quad (2.23)$$

we can compute the 0 point energy as

$$\mathcal{H} |0\rangle \propto (2\pi)^3 \delta(0) \rightarrow V \quad (2.24)$$

in a finite volume V . The zero energy is then identified as

$$E_0 = \frac{V}{2} \int \frac{d^3p}{(2\pi)^3} E_p \rightarrow \rho = \frac{E_0}{V} = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} E_p \quad (2.25)$$

as the volume diverges, the zero energy and the zero point density diverge as well. This is because $E_p = \sqrt{m^2 + p^2}$ is unbounded for large p . One way to solve this would be to insert a cutoff momentum scale, say Λ , and integrate up to the scale. In the end, we will be interested only in energy differences, so this is not a big deal. Tecnically, this divergences are called *UV divergences*.

If we want to declare that the energy of the vaccum is zero, we could order the operators, so all the destruction operators act first. This is called *normal ordering*. Then we find that quantizing fields and their functions (such as the hamiltonian) becomes an easy task: you just promote them to operators, and impose normal ordering. Given a general operator $\mathcal{O}$, we denote its normal ordering by $:\mathcal{O}:$. For example, we have $:aa^\dagger := a^\dagger a$.

Take also for example the spacial momentum p^i . Since momentum is the conserved current due to invariance with respect to spacial translations, it is given by

$$P^i = \int d^3x T^{0i} \quad (2.26)$$

Now, for a scalar field, $T^{0i} = (\partial^0\phi)(\partial^i\phi)$. So following the idea above, we make this an operator and we apply normal ordering:

$$P^i = \int d^3x : (\partial^0\phi)(\partial^i\phi) : \quad (2.27)$$

Now we have to substitute the expression of the fields in terms of ladder operators, and compute the derivatives. After some algebra one finds:

$$P^i = \int \frac{d^3p}{(2\pi)^3} \frac{p^i}{2} : (a_p a_p^\dagger + a_p^\dagger a_p) : \quad (2.28)$$

$$= \int \frac{d^3p}{(2\pi)^3} p^i a_p^\dagger a_p \quad (2.29)$$

what does this expression mean? Let's look closely: the operators in this expressions are P^i and the ladder operators. p^i is *not* an operator, it is just a 3-vector. It comes from the exponential expansion of the field, after differentiation. This expression basically means that we are building our theory in the right way: let's compute

$$P^i |p\rangle = \sqrt{2E_p} \int \frac{d^3p'}{(2\pi)^3} p'^i a_{p'}^\dagger a_{p'} a_p^\dagger |0\rangle \quad (2.30)$$

$$= \sqrt{2E_p} \int d^3p' \delta(p - p') a_p^\dagger |0\rangle \quad (2.31)$$

$$= \sqrt{2E_p} p^i a_p^\dagger |0\rangle \quad (2.32)$$

$$= p^i |p\rangle \quad (2.33)$$

so $|p\rangle$ is an eigenstate of the operator P^i , with eigenvalue p^i .

Osservazione:-

From the commutation relations of creation operators, the particle states $|p_1 \dots p_n\rangle$, are symmetric in the exchange of labels. Thus, the particles are *bosons* in the sense that they obey Bose-Einstein statistics.

2.3 The complex scalar field

A slight, but very useful, variation of the scalar real field is the scalar *complex* field. Let's define

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \quad \phi_1, \phi_2 \in \mathbb{R}^2 \quad (2.34)$$

if we give this complex field the lagrangian

$$\mathcal{L} = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 |\phi|^2 \quad (2.35)$$

then the two field components, ϕ_1, ϕ_2 independently satisfy the KG equation (with the same mass). Thus the general solution for ϕ is

$$\phi = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} (a_p e^{-ipx} + b_p^* e^{ipx}) \quad (2.36)$$

the reader might wonder why we introduce such a field, if it is just a compact way of writing two independent fields. It turns out that if you watch closely the lagrangian, you'll find a new symmetry: a global phase. If we take

$$\phi(x) \rightarrow e^{i\theta} \phi(x) \quad (2.37)$$

then the lagrangian is invariant. We then speak of a *global* $U(1)$ symmetry, at witch, via Noether's theorem, is associated a conserved charge.

Let us compute the conserved current. If we take an infinitesimal transformation, we have

$$\phi(x) \rightarrow \phi(x) + i\theta\phi(x) = \phi(x) + \theta\delta(\phi(x)) \implies \phi^*(x) \rightarrow \phi^*(x) - i\theta\phi^*(x) = \phi^*(x) + \theta\delta(\phi^*(x)) \quad (2.38)$$

so the conserved current is given by

$$j^\mu = - \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \delta(\phi^*) \right] \quad (2.39)$$

$$= -i((\partial^\mu \phi^*)\phi + (\partial^\mu \phi)\phi^*) \quad (2.40)$$

$$= i\phi^* \overleftrightarrow{\partial}^\mu \phi \quad (2.41)$$

then the charge is

$$Q = \int d^3x j^0 = i \int d^3x \phi^* \overleftrightarrow{\partial}^0 \phi = \langle \phi^* | \phi \rangle \quad (2.42)$$

When we quantize this complex field, in the usual way by simply promote the fourier coefficients to ladder operators, this time we have to pairs of ladder operators, $a_p, a_p^\dagger$ and $b_p, b_p^\dagger$. They create particle states acting on the vacuum state (witch this time is defined to satisfy $a_p |0\rangle = 0$ and $b_p |0\rangle = 0$), and they create particle states with the same mass, and spin 0. Their difference is understood in terms of the $U(1)$ charge: if one computes Q substituting the expression of the field in terms of the ladder operators, one finds

$$Q = \int d^3x j^0 = \dots = \int \frac{d^3p}{(2\pi)^3} (a_p^\dagger a_p - b_p^\dagger b_p) \quad (2.43)$$

witch is the number operators of particles created from $a_p^\dagger$, minus the number operator of particle created from $b_p^\dagger$. For this reason, the second type of particle is called *antiparticle*.

3.1 Quick review of classical electrodynamics

We'd now like to quantize a vector field, say A^μ . This, of course, will give us a quantum theory of electromagnetism. To begin with, let's then refresh our mind on how we can think of electromagnetism as a classical field theory.

Starting from a vector potential (also called gauge potential) $A^\mu = (\phi, \vec{A})$, one defines the *field strength* $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. This tensor is antisymmetric, and thus has 6 free parameters, that correspond to the electric and magnetic field. With a direct computation, one can get the relations

$$F^{0i} = -E^i, \quad F^{ij} = -\epsilon^{ijk} B_k \quad (3.1)$$

Far from sources, the lagrangian that correctly reproduces the Maxwell equation is

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \implies \partial_\mu F^{\mu\nu} = 0 \quad (3.2)$$

One of the remarkable things about electromagnetism is that it is a *gauge theory*: the vector potential A^μ is not defined uniquely, as one may take the *gauge transformation*

$$A^\mu \rightarrow A^\mu + \partial^\mu \chi \quad (3.3)$$

and the fields would not change. This can bring some practical problems as one tries to quantize the theory. Thus, when we will try to quantize this theory, we will *fix* the gauge, that is, we will select a particular gauge choice, and stick with it.

One useful gauge choice is the Coulomb (or radiative) gauge. In such, one takes $A_0 = 0, \nabla \cdot \vec{A} = 0$. The e.o.m. then become

$$\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = 0 \rightarrow \square A^\nu = 0 \quad (3.4)$$

that is, basically, a vector version of the KG equation with zero mass. The solution is then similar to the solution of the KG equation, with the mass-shell relation that now becomes $p^2 = 0$, with a *polarization* vector $\vec{\epsilon}$:

$$\vec{A} = \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_p}} \sum_{\lambda=1,2} \left[\vec{\epsilon}(\vec{p}, \lambda) a_{p,\lambda} e^{-ipx} + \vec{\epsilon}^*(\vec{p}, \lambda) a_{p,\lambda}^\dagger e^{ipx} \right], \quad \omega_p = p^0 = |\vec{p}| \quad (3.5)$$

you might wonder why we've put the polarization vector to depend only on two discrete variables, $\lambda = 1, 2$. A priori, the polarization vector could point in any direction in space, and then we should take $\lambda = 1, 2, 3$. The reason is that the condition $\vec{\nabla} \cdot \vec{A} = 0$ implies $\vec{p} \cdot \vec{\epsilon} = 0$. So $\vec{\epsilon}$ must lie in the trasverse plane, and can only point in two different directions.

If you're confused, a tipical choice of $\epsilon(\vec{k}, \lambda)$ is given by, if $\vec{k} = (0, 0, k)$, $\vec{\epsilon}(k, 1) = (1, 0, 0)$, $\vec{\epsilon}(k, 2) = (0, 1, 0)$.

3.2 Canonical quantisation

Now that we have the general solution, in a preferred gauge, you can probably guess how we're going to quantize this field. Let's promote the fourier coefficients to operators, with commutation relation

$$[a_{p,\lambda}, a_{q,\nu}^\dagger] = (2\pi)^3 \delta(p - q) \delta_{\lambda\nu} \quad (3.6)$$

The reason for doing this is that in this way we can build the fock space, exactly how we build it for the scalar field. But what does this imply for the fields in this case? compute the coniugate momentum:

$$\begin{cases} \Pi_0 = \frac{\delta}{\delta(\partial_0 A^0)} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) = 0 \\ \Pi_i = \frac{\delta}{\delta(\partial_0 A^i)} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) = \dots = -F^{0i} = E^i \end{cases} \quad (3.7)$$

then the commutation relation is

$$[A^i(t, \vec{x}), \Pi^j(t, \vec{y})] = -i \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k} \cdot (\vec{x} - \vec{y})} \frac{1}{2} \sum_{\lambda} (\epsilon^i \epsilon^{*j} + \epsilon^{*i} \epsilon^j) \quad (3.8)$$

$$= -i \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k} \cdot (\vec{x} - \vec{y})} \left(\delta^{ij} - \frac{k^i k^j}{k^2} \right) \quad (3.9)$$

If it weren't for the last term, this would be the old Dirac's delta, exactly like the scalar field. The additional term is given by our particular gauge choice. This can be verified by taking the divergence of the field in the commutator: for the gauge condition, clearly $[\vec{\nabla} \cdot \vec{A}, \Pi^j] = 0$ should hold. It is easy to check that the RHS vanishes in this case only for the additional therm. On the other end, for the same reason we find that $[\vec{A}, \vec{\nabla} \cdot \vec{E}] = 0$ (recall that the spacial coniugate momentum is just the electric field). This means that the operator obtained by taking the divergence of the electric field commutes with everything.

The classic energy momentum tensor can be computed,

$$T^{\mu\nu} = -F^{\mu\rho} \partial^\nu A_\rho + \frac{1}{4} g^{\mu\nu} F^2 \quad (3.10)$$

And from this the classical hamiltonian,

$$H = \int d^3x T^{00} = \dots = \frac{1}{2} \int d^3x (E^2 + B^2) \quad (3.11)$$

the quantum version is given by imposing normal ordering:

$$H = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \omega_k \sum_{\lambda} a_{k,\lambda}^\dagger a_{k,\lambda} \quad (3.12)$$

3.3 Spin operator and its eigenstates

Of course, the new exciting thing about the electromagnetic field is that it carries a spin. Given a particle state, how can we compute its spin? First things first, we must determine the angular momentum operator, using Noether's theorem.

The angular momentum operator is the Noether charge associated with the global rotation symmetry of the lagrangian. In formulas, we write

$$J^{jk} = \int d^3x j^{0(jk)}, \quad j^{\mu(jk)} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A^i)} (a^{\nu(jk)} \partial_\nu A^i - F^{i(jk)}) - a^{\mu(jk)} \mathcal{L} \quad (3.13)$$

Recall the meaning of all this symbols: the vector $a^{\nu(jk)}$ measures of the coordinates transform under a spacial rotation. Similarly, the factor $F^{i(jk)}$ measures how the fields transform under a spacial rotation. Since both are true vectors under rotations, we get

$$\begin{cases} a^{\ell(jk)} = \delta^{\ell j} x^k - \delta^{\ell k} x^j, & a^{0(jk)} = 0 \\ F^{i(jk)} = \delta^{ij} A^k - \delta^{ik} A^j \end{cases} \quad (3.14)$$

the conserved current is then

$$j^{0(jk)} = -(\partial_0 A^i)(x^k \partial^j - x^j \partial^k) A_i - (A^k \partial_0 A^j - A^j \partial_0 A^k) \quad (3.15)$$

Thus, the total charge is a sum of two distinct operators,

$$J^{jk} = M^{jk} + S^{jk}, \quad \begin{cases} M^{jk} = \langle A^i | L^{ij} | A^i \rangle, & L^{ij} = i(x^j \partial^k - x^k \partial^j) \\ S^{ij} = \int d^3x : (A^i \partial_0 A^j - A^j \partial_0 A^i) : \end{cases} \quad (3.16)$$

we recognize the expectation value of the angular momentum operator, that appeared also in the scalar field case. Then, we have another operator, that we then assume to be the spin operators. We'd then like to see how this operator act on the particle states of our fock space. First, let's plug in the expansion of our field in the spin operator, performing normal ordering: one finds then

$$S^{ij} = i \int \frac{d^3q}{(2\pi)^3} \sum_{\lambda\lambda'} [\epsilon^i(q, \lambda') \epsilon^{*j}(q, \lambda) - \epsilon^{*i}(q, \lambda) \epsilon^j(q, \lambda')] a_{q,\lambda}^\dagger a_{q,\lambda'} \quad (3.17)$$

we're now ready to see what

$$S^{ij} a_{k,\lambda''}^\dagger |0\rangle \quad (3.18)$$

gives. To do this, we use our usual tricks on commuting the ladder operators:

$$S^{ij} a_{k,\lambda''}^\dagger |0\rangle = i \sum_{\lambda} (\epsilon^i(k, \lambda'') \epsilon^{*j}(k, \lambda) - \epsilon^{*i}(k, \lambda) \epsilon^j(k, \lambda'')) a_{k,\lambda}^\dagger |0\rangle \quad (3.19)$$

since now we haven't said anything about the direction of k and ϵ . Take $\vec{k} = (0, 0, k)$ and let's check what the z component of the spin operator is (that is, how S^{12} acts on a particle state with momentum $\parallel \hat{z}$). We also choose polarization vectors like

$$\epsilon(k, 1) = (0, 1, 0), \epsilon(k, 2) = (1, 0, 0) \quad (3.20)$$

or, in more compact form, $\epsilon^i(k, \lambda) = \delta_\lambda^i$. Then it's easy to check that

$$S^{12} a_{k,1}^\dagger |0\rangle = i a_{k,2}^\dagger |0\rangle \quad (3.21)$$

$$S^{12} a_{k,2}^\dagger |0\rangle = -i a_{k,1}^\dagger |0\rangle \quad (3.22)$$

so we find that this states with linear polarization *are not* eigenstates of the spin operator in the longitudinal direction. But it's not hard to see that if we instead define

$$a_\pm^\dagger := \frac{1}{\sqrt{2}}(a_{k,1}^\dagger \pm i a_{k,2}^\dagger) \implies \begin{cases} S^{12} a_{k+}^\dagger = a_{k+}^\dagger |0\rangle \\ S^{12} a_{k-}^\dagger = -a_{k-}^\dagger |0\rangle \end{cases} \quad (3.23)$$

so we have explicitly built the eigenstates of the spin operator in the longitudinal direction, and we have also shown that it's eigenvalues are ± 1 . This is why we say that the photon has spin 1. Also, the eigenstates of spin are particles that are polarized circularly.

3.4 Covariant quantisation

There is another way of quantizing electrodynamics, witch doesn't require picking up a gauge that doesn't brake covariance. Of course, the naive way of quantizing the field would be to compute it's coniugate momentum,

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A^\mu)} \quad (3.24)$$

and than require that the commutation relation

$$[A^\mu(t, \vec{x}), \Pi^\nu(t, \vec{y})] = (2\pi)^3 \delta(\vec{x} - \vec{y}) g^{\mu\nu} \quad (3.25)$$

holds. But this soon one comes to a huge problem: given the free electromagnetic lagrangian, one finds $\Pi^0 = 0$, as $F^{00} = 0$. Then, $[A^0, \Pi^0] = 0$ everywhere, in contrast to the commutation relation that we were trying to implement. The theory, formulated this way, is non consistent. The trick is to choose a particular gauge, the lorentz gauge $\partial_\mu A^\mu = 0$, only at the *operator level*: we will not fix the gauge now, but after we quantize the theory, we'll only keep a subspace of states in the hilbert space, the *physical states*, that obey $\langle \text{phys} | \partial_\mu A^\mu | \text{phys} \rangle$.

What is the advantedge of doing this? Now we can take a *physically equivalent* lagrangian given by

$$\mathcal{L}' = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} (\partial_\mu A^\mu)^2 \quad (3.26)$$

where, of course, physically equivalent here does not mean that the two lagrangian give the same physics (in fact, the physics is much different as one has a $U(1)$ gauge symmetry, whereas the other does not), but it means that

taken as an operator, it is equal to the previous standard lagrangian, when acting on the states that we defined to be physical.

The cool thing about the primed lagrangian is that its equation of motion are given by

$$\square A^\nu = 0 \quad (3.27)$$

that is, massless KG. The most general solution is then

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{\lambda=0}^3 [\epsilon_\mu(p, \lambda) a_{p,\lambda} e^{-ipx} + \epsilon_\mu^*(p, \lambda) a_{p,\lambda}^* e^{ipx}] \quad (3.28)$$

This is similar to the result we had when we chose the polarization gauge. However, there is a big difference in the fact that now the polarization can point in any direction of the spacetime, so its discrete label λ now takes values from 0 to 3, instead of only 1 and 2. Thus we would try to quantize the theory by imposing the commutation relation on the ladder operators:

$$[a_{p,\lambda}, a_{q,\lambda'}^\dagger] = -(2\pi)^3 \delta(p - q) g_{\lambda\lambda'} \quad (3.29)$$

The sign minus is needed to recover the previous relation in the radiation gauge, if the labels take the 1 and 2 values. If we compute

$$\langle p, \lambda' | p, \lambda \rangle = 2\omega_p \langle 0 | a_{p,\lambda} a_{p,\lambda'}^\dagger \quad (3.30)$$

$$= -2\omega_p V g_{\lambda\lambda'} \quad (3.31)$$

so the states have *negative norm* if $\lambda = \lambda' = 0$. This would be a problem, but we must also select the physical states.

Let's then select the physical subspace. The condition for the states to be physical is equivalent to

$$(\partial_\mu A^\mu)^+ |\text{phys}\rangle = 0, \text{ if } (\partial_\mu A^\mu)^+ = -i \int \frac{d^3p}{(2\pi)^2 \sqrt{2\omega_p}} \sum_\lambda p_\mu \epsilon^\mu(p, \lambda) e^{-ipx} a_{p,\lambda} \quad (3.32)$$

We take a general particle state to be

$$|\psi\rangle = \sum_{\lambda=0}^3 c_\lambda a_{k,\lambda}^\dagger |0\rangle \quad (3.33)$$

If we impose it to be physical, a quick computation shows that $c_0 + c_3 = 0$ must hold. This is good news because it tells us that the previous states with negative norm are not physical states. There is still though a subset of states that are physical, but didn't show up in the previous quantisation. Those are the states

$$|\phi\rangle = c(a_{k,0}^\dagger - a_{k,3}^\dagger) |0\rangle, \quad c \in \mathbb{R} \quad (3.34)$$

However, a simple computation shows that they have zero norm, and their scalar product with a particle state like $|\psi_T\rangle = (c_1 a_{k,1}^\dagger + c_2 a_{k,2}^\dagger)$ is zero. So they completely decouple from the spectrum of the hamiltonian, and no experiment can detect if a state is $|\psi_T\rangle$ or $|\psi_T\rangle + |\phi\rangle$, so we can ignore them.

4

Spin 1/2 fields

4.1 Charge conjugation

Let's consider a right handed Weyl spinor. Under a Lorentz transformation, we know that this object transforms as

$$\psi_R \rightarrow \Lambda_R \psi_R, \Lambda_R = \exp \left((-i\vec{\theta} + \vec{\eta}) \cdot \frac{\vec{\sigma}}{2} \right) \quad (4.1)$$

Now consider the object $\sigma^2 \Lambda_L^* \sigma^2$. Writing the exponential as a series, and using the fact that $\sigma^2 \sigma^2 = \mathcal{I}$ and also $\sigma^2 \sigma^i \sigma^2 = -(\sigma^i)^*$ one finds

$$\sigma^2 \Lambda_L^* \sigma^2 = \Lambda_R \quad (4.2)$$

so the object $\sigma^2 \psi_L^*$ under a Lorentz transformation is mapped in

$$(\sigma^2 \psi_L^*) \rightarrow \Lambda_R \sigma^2 \psi_L^* \quad (4.3)$$

Guided by this observations, we define the *charge conjugation* on LH spinors as

$$\psi_L^c = i\sigma^2 \psi_L^* = \xi_R \quad (4.4)$$

note that, as σ^2 is a purely complex matrix, $i\sigma^2$ is real, and than

$$\xi_R^* = i\sigma^2 \psi_L \implies \psi_L = -i\sigma^2 \xi_R^* \quad (4.5)$$

so we define charge conjugation for RH Weyl spinors as

$$\psi_R^c = -i\sigma^2 \psi_R^* \quad (4.6)$$

With this definitions, it's easy to check that $(\psi_{L/R}^c)^c = \psi_{L/R}$

4.2 A lagrangian for Weyl spinors

Now, we'd like to build a lagrangian for the dynamics of Weyl spinors. To have equation of motion in the usual sense, the lagrangian must include the derivative of our spinors. On the other end, to have a lagrangian that is Lorentz invariant, we also need to have a vector to contract the derivative of the spinor. One possible way to do this is introduce the 4-vector of pauli matrices, $\sigma^\mu = (\mathbb{I}, \vec{\sigma})$ as well as $\bar{\sigma}^\mu = (\mathbb{I}, -\vec{\sigma})$. Then the objects

$$\xi_R^\dagger \sigma^\mu \psi_R, \quad \xi_L^\dagger \bar{\sigma}^\mu \psi_R \quad (4.7)$$

are 4 vectors under the Lorentz group. Thus, a possible lagrangian is given by

$$\mathcal{L} = i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L \quad (4.8)$$

this leads to the e.o.m.

$$\bar{\sigma}^\mu \partial_\mu \psi_L = 0 \implies \partial_0 \psi_L = \sigma^i \partial_i \psi_L \quad (4.9)$$

if we take another time derivative,

$$\square \psi_L = 0 \quad (4.10)$$

as $\sigma^i \sigma^j = \delta^{ij} + \epsilon^{ijk} \sigma^k$. But this is the KG massless equation! Thus we can write a general solution as

$$\psi_L(x) = u_L e^{-ipx} \quad (4.11)$$

where u_L is a constant 2-component spinor. If we put this solution again in the equation, it tells us that

$$\frac{\vec{p} \cdot \vec{\sigma}}{E} u_L = -u_L \quad (4.12)$$

where E is just the zero component of p^μ . But since the object satisfies the KG *massless* equation, we know that $E = |\vec{p}|$. Also recalling that for spinors $\vec{\sigma}$ is related to the angular momentum via

$$\vec{J} = \frac{\vec{\sigma}}{2} \quad (4.13)$$

we get in the end

$$(\hat{p} \cdot \vec{J}) u_L = -\frac{1}{2} u_L \quad (4.14)$$

we recognize the classical helicity expression on the LHS, acting on the spinor. So we find that massless LH spinors are eigenstates of helicity, with eigenvalue $-1/2$.

We can also compute the energy momentum tensor,

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_L)} (\partial^\nu \psi_L) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_L^\dagger)} (\partial^\nu \psi_L^\dagger) - g^{\mu\nu} \mathcal{L} = i\psi_L^\dagger \bar{\sigma}^\mu \partial^\nu \psi_L \quad (4.15)$$

so for example the hamiltonian density is given by

$$\mathcal{H} = T^{00} = i\psi_L^\dagger \partial_0 \psi_L \quad (4.16)$$

The lagrangian also has a *global* $U(1)$ symmetry, (that is, under $\psi_L \rightarrow e^{i\theta} \psi_L$) that leads to a conserved current:

$$j^\mu = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_L)} F_i = \psi_L^\dagger \bar{\sigma}^\mu \psi_L \implies Q = \int d^3x \psi_L^\dagger \psi_L \quad (4.17)$$

To evaluate the current, we used the fact that F_i measures the variation of the field under a infinitesimal transformation (so we quickly get $F_i = i\psi_L$) and the fact that the transformation does not act on the coordinates.

Of course, this whole discussion could have been made for RH spinors, with lagrangian

$$\mathcal{L} = i\psi_R^\dagger \sigma^\mu \psi_R \implies \sigma^\mu \partial_\mu \psi_R = 0 \quad (4.18)$$

all that follows is the same, except for the fact that the helicity is now $1/2$.

4.3 Dirac's lagrangian and equation

We've seen that if we want to have a theory that includes the parity operation, we must work with both right and left spinors. Thus, we'd like to write a lagrangian for the two fields, with the easiest possible interaction between them. To do this, note that

$$\Lambda_L^\dagger \Lambda_R = 1 = \Lambda_R^\dagger \Lambda_L \quad (4.19)$$

Then we find that, under a lorentz transformation,

$$\psi_L^\dagger \psi_R \rightarrow \psi_L^\dagger \Lambda_L^\dagger \Lambda_R \psi_R = \psi_L^\dagger \psi_R \quad (4.20)$$

so $\psi_L^\dagger \psi_R$ is a *lorentz scalar*. Let's then write our lagrangian as

$$\mathcal{L}_D = i(\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L) + i(\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R) - m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L) \quad (4.21)$$

under parity, $\psi_L \rightarrow \psi_R$ and $\bar{\sigma}^\mu \partial_\mu \rightarrow \sigma^\mu \partial_\mu$, so this lagrangian is parity invariant. If we compute the euler-lagrange equations for the two fields, we get

$$\begin{cases} i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R \\ i\sigma^\mu \partial_\mu \psi_R = m\psi_L \end{cases} \quad (4.22)$$

It's clear that the spinors are not anymore eigenstates of helicity. Do those equation still imply the Klein-Gordon equation? Let's check: start by applying $i\sigma^\nu\partial_\nu$ to the first equation,

$$-\bar{\sigma}^\nu\sigma^\mu\partial_\mu\partial_\nu\psi_L = mi\sigma^\nu\partial_\nu\psi_R = m^2\psi_L \quad (4.23)$$

In this way, using the second equation, we have successfully decoupled the two equations. Using the known fact that $\bar{\sigma}^\mu\sigma^\nu + \sigma^\mu\bar{\sigma}^\nu = 2g^{\mu\nu}$, we find

$$(\square + m^2)\psi_L = 0 \quad (4.24)$$

so Dirac's equation implies the KG equation, and also we can interpret the m parameter as the mass of the field. Note that there was no way of inserting a mass therm in the previous lagrangians who only contained right or left spinors, without braking lorentz invariance.

There is a way of writing the lagrangian and the equations of motion in a more compact form. In particular we consider an objects (a 4-component spinor) that includes both the left and right handed spinor. Of course there are infinite ways to do this, but if we take the *chiral representation*, we write

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad (4.25)$$

we also have to introduce a 4×4 matrix that contains both $\sigma^\mu, \bar{\sigma}^\mu$, that are called the gamma Dirac's matrixes:

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \quad (4.26)$$

Importantly, this objects satisfy the *Clifford algebra*:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \quad (4.27)$$

Using also the convention that $\not{A} = \gamma^\mu A_\mu$, then we write

$$\mathcal{L} = \bar{\Psi}(i\not{\partial} - m)\Psi \implies (i\not{\partial} - m)\Psi = 0 \quad (4.28)$$

where $\bar{\Psi} = \Psi^\dagger\gamma^0$ (sometimes called the Dirac adjoint). You can check that $\bar{\Psi}\Psi = \psi_L^\dagger\psi_R + \psi_R^\dagger\psi_L$.

It is also useful to introduce

$$\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^4 \implies \gamma^5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad (4.29)$$

That satisfies $\{\gamma^5, \gamma^\nu\} = 0$. This can be checked right away because the clifford algebra implies that γ^μ anti-commutes with γ^ν whenever $\mu \neq \nu$. With γ^5 we can define the projectors operator as

$$P_L := \frac{1 - \gamma^5}{2}, \quad P_R := \frac{1 + \gamma^5}{2} \quad (4.30)$$

so that

$$P_L \Psi = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}, \quad P_R \Psi = \begin{pmatrix} 0 \\ \psi_R \end{pmatrix} \quad (4.31)$$

Other representation can be useful in differt applications. For example the standard representation is taken as

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_R + \psi_L \\ \psi_R - \psi_L \end{pmatrix} \quad (4.32)$$

under a different representation, Ψ and γ^μ change, but is is shown that the lagrangian and the e.o.m. remain the same.

Let's now study the solutions of the Dirac's equation. Since it implies the KG equation, a general solution will be written as

$$\Psi = c_1 \Psi_1 + c_2 \Psi_2, \quad \Psi_1 = u(p) e^{-ipx}, \quad \Psi_2 = v(p) e^{ipx} \quad (4.33)$$

with u, v 4-component spinors. Dirac's equations then become

$$\begin{cases} (\not{p} + m)u(p) = 0 \\ (\not{p} - m)v(p) = 0 \end{cases} \quad (4.34)$$

if $m > 0$, we can boost ourselves in the rest frame of the particle. There, $p^\mu = (m, 0, 0, 0)$. The equation then tells us $u_L = u_R$. A convenient normalization is given by $u_L = u_R = \sqrt{m} \xi$, with ξ a 2-component spinor, such that $\xi^\dagger \xi = 1$. To find the solution in a general frame, we perform a lorentz boost on this spinor.

If we take the ultra-relativistic limit of this expression, that is, we take $p = (E, 0, 0, E)$, then we get that two possible spinor are given by

$$u^1(p) = \sqrt{2E} \begin{pmatrix} 0 \\ \xi^1 \end{pmatrix}, \quad u^2(p) = \begin{pmatrix} \xi^2 \\ 0 \end{pmatrix} \quad (4.35)$$

In a more compact expression, we can write

$$u(p) = \begin{pmatrix} \sqrt{p \cdot \vec{\sigma}} \xi \\ \sqrt{p \cdot \vec{\sigma}} \bar{\xi} \end{pmatrix} \quad (4.36)$$

Similarly for the other spinor, we get

$$v(p) = \begin{pmatrix} \sqrt{p \cdot \vec{\sigma}} \xi \\ -\sqrt{p \cdot \vec{\sigma}} \bar{\xi} \end{pmatrix} \quad (4.37)$$

Let's now take a look at the global simmetries that this classical theory possesses. To begin with, we consider

$$\Psi \rightarrow e^{i\alpha} \Psi \quad (4.38)$$

this is clearly a realized simmetry, for the Dirac's lagrangian. To see this explicitly, remember the Dirac's adjoint operation, and derive how it transforms:

$$\bar{\Psi} = \Psi^\dagger \gamma^0 \rightarrow \Psi^\dagger \gamma^0 e^{-i\alpha} = \bar{\Psi} e^{-i\alpha} \quad (4.39)$$

Since α is a constant and does not depend on the spacetime coordinates, then the lagrangian is invariant.

If instead we take

$$\Psi = e^{i\beta\gamma^5} \Psi \implies \begin{cases} \psi_L \rightarrow e^{i\beta} \psi_L \\ \psi_R \rightarrow e^{-i\beta} \psi_R \end{cases} \quad (4.40)$$

this is sometimes called a *chiral trasformation*. If we see how the trasformation act on the lagangian, we see that the term in the derivative is left invariant. However, the mass term transforms as

$$\bar{\Psi} m \Psi \rightarrow m \bar{\Psi} e^{-2i\beta\gamma^5} \Psi \quad (4.41)$$

so the mass term *brakes* the chiral symmetry.

4.4 Canonical quantisation

We're now ready to quantize the spinor field. The interesting thing is that the mosto obvious way to do it, impose the usual commutation relations, is very much the wrong approach. This would lead to a theory that doesn't posses a ground state, with the eigenvalues of the hamiltonian unbounded from below. Instead we try to quantize imposing anti-commutation relations:

$$\{\Psi_a(t, \vec{x}), \Psi_b^\dagger(t, \vec{y})\} = \delta_{ab} \delta(\vec{x} - \vec{y}) \quad (4.42)$$

Remembering the expression of a Dirac field in terms of plane waves (that we cam always make, as the Dirac euqation implies the KG equation) we get the ladder operators,

$$\{a_p^r, a_q^{s\dagger}\} = \{b_p^r, b_q^{s\dagger}\} = (2\pi)^3 \delta(\vec{p} - \vec{q}) \delta^{rs} \quad (4.43)$$

and the other anti-commutators are zero. As always, define the vacuum state as the one that gets annihilated by both a_{ps}, b_{ps} . Then particle states are obtained acting on the vaccum via creation operators.

$$a_{ps}^\dagger |0\rangle \rightarrow \text{fermion}, \quad b_{ps}^\dagger |0\rangle \rightarrow \text{anti-fermion} \quad (4.44)$$

To get the correct normalization, that is to say to get

$$\langle \vec{q}, r | \vec{p}, s \rangle = (2\pi)^3 2E_p \delta(\vec{q} - \vec{p}) \delta_{rs} \quad (4.45)$$

we define

$$|\vec{p}, s\rangle := \sqrt{2E_p} a_{\vec{p},s}^\dagger |0\rangle \quad (4.46)$$

Observe that due to the anti-commutation relation, this particle states obey Fermi-Dirac statistics.

As always, we can compute important physical operators, such as the Hamiltonian, that is classically given by

$$\mathcal{H} = i\Psi^\dagger \partial^0 \Psi - \bar{\Psi}(i\not{\partial} - m)\Psi = \bar{\Psi}(-i\gamma^i \partial_i + m)\Psi \quad (4.47)$$

we get the quantum operator by imposing normal ordering:

$$\mathcal{H} = \dots = \int \frac{d^3p}{(2\pi)^3} \sum_{s=1,2} E_p (a_{\vec{p},s}^\dagger a_{\vec{p},s} + b_{\vec{p},s}^\dagger b_{\vec{p},s}) \quad (4.48)$$

and also the same procedure gives the momentum spacial operator:

$$\vec{p} = \int \frac{d^3p}{(2\pi)^3} \vec{p} \sum_{s=1,2} (a_{\vec{p},s}^\dagger a_{\vec{p},s} + b_{\vec{p},s}^\dagger b_{\vec{p},s}) \quad (4.49)$$

The most interesting thing to explore is the spin of the particle states. For a spinor Weyl field ψ_L, ψ_R the spin operator is given by

$$S^i = i\epsilon^{ijk} S^{jk} = \frac{\sigma^i}{2} \quad (4.50)$$

So the first problem we need to solve is find an appropriate form for this operator when acting on a Dirac spinor. This is not hard, let's just define

$$\Sigma^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} \implies S^i = \frac{1}{2} \int d^3x (\Psi^\dagger \Sigma^i \Psi) \quad (4.51)$$

And if we consider a particle in its rest frame, labeled by and index s , we'll write it's state as

$$a_{0,s}^\dagger |0\rangle \quad (4.52)$$

To find how S_z acts on this state, use the fact that since there are no particles in the vacuum state, it gets annihilated by the S_z operator. So we write

$$S_z a_{0,s}^\dagger |0\rangle = [S_z, a_{0,s}^\dagger] |0\rangle \quad (4.53)$$

Now we can write S_z in terms of the ladder operators. One finds that it's only component that has a non-trivial commutation relation with $a_{0,s}^\dagger$ is given by

$$S_z = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \sum_{r,s'} a_{\vec{p},r}^\dagger a_{\vec{p},s} u^{s'\dagger}(p) \frac{\Sigma^3}{2} u^r(p) \quad (4.54)$$

In particular, the commutation relation is

$$[a_{p,r}^\dagger, a_{qs'}] = \dots = (2\pi)^3 \delta_{ss'} \delta(\vec{p} - \vec{q}) \quad (4.55)$$

One then finds

$$[S_z, a_{0s}^\dagger] |0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \sum_{rs'} (2\pi)^3 \delta_{ss'} \delta^{(3)}(\vec{p}) u^{s'+}(p) \frac{\Sigma^3}{2} u'^r(p) |0\rangle \quad (4.56)$$

$$= \frac{1}{2m} \sum_r u^{s+}(0) \frac{\Sigma^3}{2} u^r(0) a_{0r}^\dagger |0\rangle \quad (4.57)$$

To proceed, write explicitly the spinors:

$$u^s(p=0) = \sqrt{m} \begin{pmatrix} \xi \\ \xi \end{pmatrix} \implies [S_z, a_{0s}^\dagger] |0\rangle = \frac{1}{2} \sum_r (\xi^{r+} \quad \xi^{r+}) \frac{\Sigma^3}{2} \begin{pmatrix} \xi^r \\ \xi^r \end{pmatrix} a_{0s}^\dagger |0\rangle \quad (4.58)$$

But remember that the 2-dimensional spinors are

$$\xi^{r=1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi^{r=2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4.59)$$

So the eigenvalues are $\pm 1/2$, if $s = 1, 2$.

Lastly, remember that the lagrangian has a $U(1)$ global symmetry. The charge operator, after applying normal ordering, looks like

$$Q = \int \frac{d^3p}{(2\pi)^3} \sum_{s_1, s_2} (a_{ps}^\dagger a_{ps} - b_{ps}^\dagger b_{ps}) \quad (4.60)$$

so particles (created by $a^\dagger$) carry a positive charge, while antiparticles, created by $b^\dagger$ have a negative charge.

5

Interacting Fields and perturbation theory

5.1 The S matrix

Since now, we've only considered non-interacting theories. Since we'd like to use this theory to make predictions about real events, we have to start to take into consideration what happens when we have interaction between fields.

The main problem when one introduces interaction, is that normally the equations of motion become non linear: thus we cannot expand the solutions in terms of plane waves anymore. To be more specific, we can pick a preferred time, and there expand via fourier analysis our fields: but then this will evolve in time through a non-linear Hamiltonian.

For example, a one particle state $a_k^\dagger |0\rangle$ can evolve into $(a_k^\dagger)^2 |0\rangle$, i.e. a two particles state. This means that the Hilbert space of the interacting theory is going to be very different from the hilbert space of the free theories.

We're going to assume that the interactions are *local*, and that they cointain only the fields, not their derivatives. The hamiltonian of the interaction is going to be labeled

$$H_{int} = \int d^3x \mathcal{H}_{int} = - \int d^3x \mathcal{L}_{int} \quad (5.1)$$

Some popular interaction theories are given by

1. ϕ^4 theory (auto-interacting scalar field):

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 \quad (5.2)$$

2. quantum electrodynamics (interaction between spinor and gauge $U(1)$ field):

$$\mathcal{L}_{QED} = \bar{\psi}(i\not{\partial} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\psi}\gamma^\mu A_\mu\psi \quad (5.3)$$

3. Yukawa coupling (interaction between scalar and spinor field):

$$\mathcal{L}_{YK} = \bar{\psi}(i\not{\partial} - m)\psi + \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2 - g\bar{\psi}\psi\phi^2 \quad (5.4)$$

What we're mainly interested in computing in QFT are scattering amplitudes, i.e. probabilities that a certain interaction will occur. An exact computation of this processes can be done only in $1+1$ dimensions. In a more realistic scenario, one must compute the amplitudes using perturbation theory.

So far, we've only worked in the Heisenberg picture. For a moment, let's get back to the Schroedinger picture, and consider a state $|a\rangle$ ad a fixed time T . Of course, this state will evolve in time, but at the beggining, it will be an

eigenstate of a set of operators, that we label by a . Similarly, we also consider a different state $|b\rangle$. If we let the states evolve with time, we get

$$|a(t)\rangle = e^{-iH(t-T)} |a\rangle \quad (5.5)$$

and the matrix element of the transition $|a\rangle \rightarrow |b\rangle$ at a time T_f will be

$$\langle b | e^{-iH(T_f-T)} | a \rangle \quad (5.6)$$

We define the S matrix as the matrix whose element are given by the latter amplitude, as the difference in time becomes infinite:

$$\langle b | S | a \rangle := \lim_{(T_f-T) \rightarrow \infty} \langle b | e^{-iH(T_f-T)} | a \rangle \quad (5.7)$$

Claim 5.1.1

From probability conservation, $SS^\dagger = 1$, so S is *unitary*.

To see this, define a complete set of states $|n\rangle$ such that $\sum_n |n\rangle \langle n| = 1$. Now, since the set is complete, the probability that $|a\rangle$ will transition into one of this is 1, in formulas

$$\sum_n |\langle n | S | a \rangle|^2 = 1 \implies \sum_n \langle a | S^\dagger | n \rangle \langle n | S | a \rangle = 1 \implies \langle a | SS^\dagger | a \rangle = 1 \quad (5.8)$$

And if the starting state is correctly normalized, S must be unitary.

Many times it's useful to write

$$S = \mathbb{I} + iT \quad (5.9)$$

that is, separate the probability that the state will remain the same, or having particle interaction.

Now let's get back to the Heisenberg picture: remembering how the two pictures are related, the matrix elements of the S matrix are quickly found:

$$\langle b | S | a \rangle = \langle b, T_f | a, T_i \rangle \quad (5.10)$$

Now, it's clear that we'd like a way of computing the S matrix. One way is called the LSZ formula. Consider then the matrix element between two states, of n and m particles, so $\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \vec{k}_1, \dots, \vec{k}_m, T_i \rangle$. Imagine that this particle state are created by a real scalar free field. We can invert the expansion of the field to obtain the ladder operators:

$$\sqrt{2E_k} a_k = i \int d^3x e^{ikx} \overleftrightarrow{\partial}_0 \phi_{free} \quad (5.11)$$

$$\sqrt{2E_k} a_k^\dagger = -i \int d^3x e^{-ikx} \overleftrightarrow{\partial}_0 \phi_{free} \quad (5.12)$$

If $T_f - T_i \rightarrow \infty$, particles are well separated. Assuming the interaction decreases in magnitude with distance, we can imagine that in that limit, the relevant theory is the free theory. We then write

$$\phi(x) \rightarrow \begin{cases} \sqrt{Z}\phi_{in}(x), & \text{as } t \rightarrow -\infty \\ \sqrt{Z}\phi_{out}(x), & \text{as } t \rightarrow \infty \end{cases} \quad (5.13)$$

with Z a normalizing factor. Thus, we have

$$\sqrt{2E_k}a_k^{\dagger(in)} = -i \int d^3x e^{-ikx} \overleftrightarrow{\partial}_0 \phi_{in} \quad (5.14)$$

$$= -i\sqrt{Z} \lim_{t \rightarrow -\infty} \int d^3x e^{-ikx} \overleftrightarrow{\partial}_0 \phi(x) \quad (5.15)$$

and in the same way,

$$\sqrt{2E_k} = i\sqrt{Z} \lim_{t \rightarrow \infty} \int d^3x e^{-ikx} \overleftrightarrow{\partial}_0 \phi(x) \quad (5.16)$$

Notice that the two operators act on different spaces: the initial/final states at T_i/T_f .

This gives a way to compute the S matrix element:

$$\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \vec{k}_1, \dots, \vec{k}_m, T_i \rangle = \sqrt{2E_{k_1}} \langle \vec{p}_1, \dots, \vec{p}_n, T_f | a_{k_1}^{\dagger(in)} | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.17)$$

$$= -iZ^{-1/2} \lim_{t \rightarrow -\infty} \int d^3x e^{-ik_1x} \langle \vec{p}_1, \dots, \vec{p}_n, T_f | \overleftrightarrow{\partial}_0 \phi | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.18)$$

Now this is good but doesn't quite look lorentz covariant. We can write it in a way that makes it more explicit, using the fundamental theorem of calculus:

$$\left(\lim_{t \rightarrow \infty} - \lim_{t \rightarrow -\infty} \right) f(t) = \int_{-\infty}^{+\infty} dt \frac{df}{dt} \implies \sqrt{2E_k}(a_k^{\dagger(in)} - a_k^{\dagger(out)}) = \int_{-\infty}^{+\infty} \int d^3x (iZ^{1/2}) \partial_0 (e^{-ikx} \overleftrightarrow{\partial}_0 \phi) \quad (5.19)$$

Applying the derivative and remembering that $\partial_0^2 e^{-ikx} = -k_0^2 e^{-ikx} = (\nabla^2 - m^2)e^{-ikx}$, we write

$$\sqrt{2E_k}(a_k^{\dagger(in)} - a_k^{\dagger(out)}) = iZ^{1/2} \int d^4x (e^{-ikx} \partial_0^2 \phi - \phi (\nabla^2 - m^2) e^{ikx}) \quad (5.20)$$

Now, we want to work with quantities that are localized in space: we can then integrate by parts in space, and drop the boundary terms, to get

$$\sqrt{2E_k}(a_k^{\dagger(in)} - a_k^{\dagger(out)}) = iZ^{1/2} \int d^4x e^{-ikx} (\square + m^2) \phi(x) \quad (5.21)$$

please note that the integral is *not* zero, as the field doesn't satisfy the KG equation everywhere in spacetime, but only at some fixed times.

Using this useful relation, we get back to our matrix elements and we write

$$\sqrt{2E_{k_1}} \langle \vec{p}_1, \dots, \vec{p}_n, T_f | (a_{k_1}^{\dagger(in)} - a_{k_1}^{\dagger(out)}) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle = \quad (5.22)$$

$$= iZ^{1/2} \int d^4x e^{-ikx} (\square + m^2) \langle \vec{p}_1, \dots, \vec{p}_n, T_f | \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.23)$$

this is exactly the matrix element from the beginning, as $a_{k_1}^{\dagger(out)}$ acts on the bra annihilating it.

Ultimately, we'd like to use this trick to eliminate all the states. We have to understand how to eliminate the final states, then. To do this we start with

$$\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.24)$$

and we try to remove the $\vec{p}_1$ particle, with the $a_{p_1}^{(out)}$ operator. So we get

$$\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle = \sqrt{2E_{p_1}} \langle \vec{p}_2, \dots, \vec{p}_n, T_f | a_{p_1}^{(out)} \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.25)$$

we define the *time ordered* product:

$$T \{ \phi(x) \phi(y) \} = \begin{cases} \phi(y) \phi(x) & \text{if } x^0 < y^0 \\ \phi(x) \phi(y) & \text{if } x^0 > y^0 \end{cases} \quad (5.26)$$

Since $a_p^{(out)}$ is defined as $y^0 \rightarrow \infty$, it goes always first in the time product. Similarly $a^{(in)}$ goes always last. We notice than that

$$\sqrt{2E_{p_1}} \langle \vec{p}_2, \dots, \vec{p}_n, T_f | T \{ (a_{p_1}^{(out)} - a_{p_1}^{(in)}) \phi(x) \} | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle = \quad (5.27)$$

$$= \sqrt{2E_{p_1}} \langle \vec{p}_2, \dots, \vec{p}_n, T_f | a_{p_1}^{(out)} \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.28)$$

Here we see why we had to introduce the time ordering product: since $T(a_p^{(in)} \phi(x)) = \phi(x) a_p^{(in)}$, the destruction operator can act on the initial state and annihilate it. As before we get

$$\sqrt{2E_{p_1}} (a_{p_1}^{(out)} - a_{p_1}^{(in)}) = -iZ^{-1/2} \int d^4x e^{ip_1 y} (\square + m^2) \phi(y) \quad (5.29)$$

in the end, we get

$$\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \phi(x) | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle = iZ^{1/2} \int d^4y e^{ip_1 y} (\square + m^2) \langle \vec{p}_2, \dots, \vec{p}_n, T_f | T \{ \phi(y) \phi(x) \} | \vec{k}_2, \dots, \vec{k}_m, T_i \rangle \quad (5.30)$$

This procedure can now be easily iterated, and we find

$$\langle \vec{p}_1, \dots, \vec{p}_n, T_f | \vec{k}_1, \dots, \vec{k}_m, T_i \rangle = \quad (5.31)$$

$$= \int \prod_i^m d^4 x^i \prod_j^n d^4 y^j \exp i \sum_j^n p_j y_j - i \sum_i^m k_i x_i (\square_{x_i} + m^2) \dots (\square_{y_1} + m^2) \dots \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle \quad (5.32)$$

where with $|\Omega\rangle$ we've labeled the vacuum of the interacting theory. In the Schroedinger's picture, the last equation tells us

$$\langle \vec{p}_1, \dots, \vec{p}_n | iT | \vec{k}_1, \dots, \vec{k}_m \rangle \quad (5.33)$$

since the states are supposed to be different from one another. We understand then that the quantity

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle \quad (5.34)$$

this objects are called *green functions* (for reason that'll become apparent later on), or time-ordered correlation functions. Tipicalli one deals with this functions under fourier transform: this is the natural thing to do, as we've been labelling all the states by their momenta. Another reason why this is useful is that the action of the KG operators becomes straightforward:

$$G(x_1, \dots, x_n) = \int \prod_i^n \frac{d^4 k_i}{(2\pi)^4} e^{-i \sum_i k_i x_i} \tilde{G}(k_1, \dots, k_n) \implies (\square_{x_j} + m^2)G = - \int \prod_i^n \frac{d^4 k_i}{(2\pi)^4} (k_j^2 - m^2) e^{-i \sum_i k_i x_i} \tilde{G} \quad (5.35)$$

Putting together what we've learned, we get the famous LSZ formula:

$$\int \prod_i^m d^4 x^i e^{-i k_i x_i} \int \prod_j^m d^4 y^j e^{i p_j y_j} \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \quad (5.36)$$

$$\left(\prod_i^m \frac{i\sqrt{Z}}{k_i^2 - m^2} \right) \left(\prod_j^n \frac{i\sqrt{Z}}{p_j^2 - m^2} \right) \langle \vec{p}_1, \dots, \vec{p}_n | iT | \vec{k}_1, \dots, \vec{k}_m \rangle \quad (5.37)$$

Notice that the RHS is singular when the momenta are on-shell. Of course, this means that the green's functions (that appear on the LHS) are singular too. In practice, we can use the RHS to read the residues of the poles of the green functions.

5.2 Perturbative expansion of corelators

The time product in the LSZ formula is expressed in terms of the interacting fields. This is not very useful in practice. Instead, we can switch to the interaction picture. Avoiding for the moment the details of the procedure, one finds in the end

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_N) \} | \Omega \rangle = \frac{\langle 0 | T \{ \phi_I(x_1) \dots \phi_I(x_N) e^{-i \int d^4 x \mathcal{H}_I} \} | 0 \rangle}{\langle 0 | T \{ e^{-i \int d^4 x \mathcal{H}_I} \} | 0 \rangle} \quad (5.38)$$

where ϕ_I in this equations are the interaction picture fields. Still, we have to understand, or develop a procedure, to compute this time ordered products.

To get an idea on how we might tackle this problem, we look at a very meaningful example:

Esempio 5.2.1 (The Feynman propagator)

Consider the vacuum expectation value of the time ordered product of two fields,

$$\langle 0 | T \{ \phi_I(x) \phi_I(y) \} | 0 \rangle \quad (5.39)$$

this is called a Feynman propagator, or sometimes two point function (Also the green function, as it turns out that it's also the green function of the KG equation). We are going to compute this for a real scalar field.

Write

$$\phi = \phi^+ + \phi^-, \quad \begin{cases} \phi^+ = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} a_p e^{-ipx} \\ \phi^- = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} a_p^\dagger e^{ipx} \end{cases} \quad (5.40)$$

This is useful as $\phi^+ |0\rangle = \langle 0 | \phi^- = 0$.

For now, assume $x^0 > y^0$:

$$T \{ \phi(x) \phi(y) \} = \phi(x) \phi(y) = \phi^+(x) \phi^+(y) + \phi^+(x) \phi^-(y) + \phi^-(x) \phi^+(y) + \phi^-(x) \phi^-(y) \quad (5.41)$$

It would be very useful if we managed to write this in terms of a normal ordered term, plus some rest. This is because we are going to look at the expectation value of this operator on the vacuum state. We see that the expression above is almost time ordered, except for

$$\phi^-(x) \phi^+(y) = \phi^+(y) \phi^-(x) + [\phi^-(x), \phi^+(y)] \implies T \{ \phi(x) \phi(y) \} = : \phi(x) \phi(y) : + [\phi^+(x), \phi^-(y)] \quad (5.42)$$

A similar computation can be performed if $x^0 < y^0$ (Just swap $x \leftrightarrow y$). In the end one finds that, in general,

$$T \{ \phi(x) \phi(y) \} = : \phi(x) \phi(y) : + \theta(x^0 - y^0) [\phi^+(x), \phi^-(y)] + \theta(y^0 - x^0) [\phi^+(y), \phi^-(x)] \quad (5.43)$$

So the Feynman propagator is given by

$$\langle 0 | \theta(x^0 - y^0) [\phi^+(x), \phi^-(y)] + \theta(y^0 - x^0) [\phi^+(y), \phi^-(x)] | 0 \rangle = \quad (5.44)$$

$$= \theta(x^0 - y^0) [\phi^+(x), \phi^-(y)] + \theta(y^0 - x^0) [\phi^+(y), \phi^-(x)] = D(x - y) \quad (5.45)$$

As it's clear that the commutator of the two fields is a number (not an operator anymore), and we can assume the vacuum to be normalized.

Writing explicitly the two fields, and using the commutation relation $[a_p, a_{p'}^\dagger] = (2\pi)^3 \delta(p - p')$, we quickly find

$$D(x - y) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left(\theta(x^0 - y^0) e^{-ip(x-y)} + \theta(y^0 - x^0) e^{ip(x-y)} \right) \quad (5.46)$$

The expression is more elegant if we put it in its covariant form:

$$D(x - y) = \int \frac{d^4p}{(2\pi)^4} \frac{i e^{-ip(x-y)}}{p^2 - m^2 + i\epsilon} \quad (5.47)$$

The equivalence of the two can be found starting from the last one, separating time and spatial components of the momentum, and then compute the time integral using the residue theorem.

5.2.1 Wick's theorem

We've seen that the computation of the 2-point function becomes straightforward if we convert the time ordered operator into a normal ordered expression plus a rest. How does this generalize when we want to compute the n-point functions?

The answer is given by Wick's theorem:

$$T \{ \phi(x_1) \dots \phi(x_N) \} = : \phi(x_1) \dots \phi(x_N) : + (\text{All possible combinations of time ordering and contractions}) \quad (5.48)$$

Where by contraction of two fields we mean the Feynman propagator between the two, $D(x_i - x_j)$.

Wick's theorem can be proven by induction for every N . Of course, we've already proved the base of induction, in computing the 2-point function. Since all of this might start to feel a bit abstract, we will now perform a simple computation, of the matrix element of a process using both the LSZ formula, and Wick's theorem.

Esempio 5.2.2 (2 → 2 scattering amplitudes in ϕ^4 theory)

Consider a theory with lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) + \frac{\lambda}{4!} \phi^4 \quad (5.49)$$

this describes an auto-interacting scalar field, with a 4-vertex. What does this mean? Let's consider the following process:

$$\phi(p_1) + \phi(p_2) \rightarrow \phi(k_1) + \phi(k_2) \quad (5.50)$$

write explicitly the LSZ formula:

$$\int d^4x_1 d^4x_2 d^4y_1 d^4y_2 e^{i(p_1x_1 - p_2x_2 - k_1y_1 - k_2y_2)} \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(y_1) \phi(y_2) \} | \Omega \rangle = \quad (5.51)$$

$$= \left(\frac{1}{p_1^2 - m^2} \right) \left(\frac{1}{p_2^2 - m^2} \right) \left(\frac{1}{k_1^2 - m^2} \right) \left(\frac{1}{k_2^2 - m^2} \right) \langle p_1, p_2 | iT | k_1 k_2 \rangle \quad (5.52)$$

6

The path integral formulation of QFT

Until now, we've only worked in the canonical quantisation. We've also come pretty far with it, as we managed to describe free theories, interacting theories (although only perturbatively), computing also cross-sections. Why do we want to go back to the roots and introduce another quantisation procedure? There is more than one reason:

1. some theories have strong coupling (e.g. QCD at hadronic energies), and cannot be described using perturbation theory. In the path integral formalism, one can treat interacting fields exactly.
2. we've seen that gauge invariance led to some problems when quantizing a gauge theory, even in the abelian case (electrodynamics). This gets worse if one considers a non-abelian gauge theory, and path integral formalism can avoid this problem.
3. The formalism highlights a profound connection between QFT and statistical mechanics.

6.1 Formalism in standard QM

In the Schroedinger picture, the amplitude of a state $|q_i\rangle$ going into a state $|q_f\rangle$ after a time ΔT is given by

$$A = \langle q_f | e^{-iH\Delta T} | q_i \rangle \quad (6.1)$$

Instead, in the Heisenberg picture we have

$$A = \langle q_f, T_f | q_i, T_i \rangle \quad (6.2)$$

note that in the Heisenberg picture the states are time independent, and the operators are the objects that evolve with time. Still, we denote a state with a time label, to indicate that at that time, the state is known to be the eigenstate of a certain set of operators, with some eigenvalues.

At every time, the states $|q_j, T_j\rangle$ form a complete set, so we could write

$$A = \int dq_j \langle q_f, T_f | q_j, T_j \rangle \langle q_j, T_j | q_i, T_i \rangle \quad (6.3)$$

But also

$$A = \int dq_1 \dots dq_{N-1} \langle q_f, T_f | q_1, T_1 \rangle \dots \langle q_{N-1}, T_{N-1} | q_i, T_i \rangle \quad (6.4)$$

for notational purposes, write $q_i = q_0$ and $q_f = q_N$ and similarly for T . We then have for the amplitude

$$A = \int dq_1 \dots dq_{N-1} \prod_{m=0}^{N-1} \langle q_m T_m | q_{m+1} T_{m+1} \rangle \quad (6.5)$$

How do we write this is the Schroedinger picture? Recall the "conversion relation" between the two pictures:

$$\langle q_m T_m | q_{m+1} T_{m+1} \rangle = \langle q_m | e^{-iH(T_{m+1}-T_m)} | q_{m+1} \rangle \quad (6.6)$$

Suppose T_i are all evenly spaced, say of ϵ , than we write

$$\langle q_m | e^{-iH\epsilon} | q_{m+1} \rangle \quad (6.7)$$

We can also introduce an integration over the momentum states:

$$\langle q_{m+1} T_{m+1} | q_m T_m \rangle = \int dp_m \langle q_{m+1} | p_m \rangle \langle p_m | e^{-iH\epsilon} | q_m \rangle \quad (6.8)$$

This is convenient because $H = H(p, q)$ and we can either act on the position ket or on the momentum bra, and in both cases we work with eigenstates. This is "legal" as we're summing over all momentum.

If we normalize $\langle q, p \rangle = e^{ipq}$, then

$$\langle q_{m+1} T_{m+1} | q_m T_m \rangle = \int dp_m e^{iq_{m+1}p_m} \langle p_m | e^{-iH(p_m, q_m)} | q_m \rangle \quad (6.9)$$

notice that now H is a function of the eigenvalue, not an operator anymore.

Finally, we write all as a big exponential:

$$\langle q_{m+1} T_{m+1} | q_m T_m \rangle = \int dp_m \exp \left\{ -i\epsilon \left(H(p_m, q_m) - \frac{q_{m+1} - q_m}{\epsilon} p_m \right) \right\} \quad (6.10)$$

If we do this for every therm in the amplitude A , we get

$$A = \int dq_1 dp_1 \dots dq_{N-1} dp_{N-1} \exp \left\{ -i\epsilon \sum_{m=0}^{N-1} \left(H(p_m, q_m) - \frac{q_{m+1} - q_m}{\epsilon} p_m \right) \right\} \quad (6.11)$$

Now perform the limit when $\epsilon \rightarrow 0$: this means we're integrating over all possible paths $q(t)$, with fixed boundary conditions, i.e. $q(T_i) = q_i, q(T_f) = q_f$. Defining $[dpdq] = \lim_{\epsilon \rightarrow 0} dq_1 dp_1 \dots dq_{N-1} dp_{N-1}$, we get:

$$A = \int_{q(T_i)=q_i}^{q(T_f)=q_f} [dpdq] \exp \left\{ i \int_{T_i}^{T_f} dt (p\dot{q} - H) \right\} \quad (6.12)$$

Normally, the integral over p is gaussian (from the kinetic part of H), and can be performed, leaving just a normalization that we put in $[dq]$. We also recognize the lagrangian in the integrand, so the integral is actually the classical action:

$$A = \int [dq] \exp \left\{ \frac{i}{\hbar} S \right\} \quad (6.13)$$

The classical limit is easily checked as $\hbar \rightarrow 0$: if we consider a small change in the trajectory, $q(t) \rightarrow q(t) + \delta q(t)$, this give large oscillations in the integrand, that average to zero. So the only contribution comes from the particular path that leaves the action unchanged under small oscillations, i.e. we recover the minimal action principle of classical mechanics.

Consider now what happens if in the integrand of the path integral we insert a factor $q(t)$, with $t \in [T_i, T_f]$:

$$\int [dq] q(t) e^{iS} \quad (6.14)$$

we can always write

$$\int [dq] = \int d\bar{q} \int_{q(T_i)=q_i}^{q(t)=\bar{q}} [dq] \int_{q(t)=\bar{q}}^{q(T_f)=q_f} [dq] \quad (6.15)$$

Split also the action integral to split the integrals:

$$\int d\bar{q} \left(\int_{q(T_i)=q_i}^{q(t)=\bar{q}} [dq] e^{i \int_{T_i}^t L} \right) \bar{q} \left(\int_{q(t)=\bar{q}}^{q(T_f)=q_f} [dq] e^{i \int_t^{T_f} L} \right) \quad (6.16)$$

we recognize the terms in the parenthesis to be just amplitudes,

$$\int d\bar{q} \langle q_i T_i | \bar{q} t \rangle \bar{q}(t) \langle \bar{q} t | q_f T_f \rangle = \langle q_i T_i | q(t) | q_f T_f \rangle \quad (6.17)$$

where we used the completeness relation.

Importantly, the generalizations for this for an arbitrary number of functions is very useful for QFT. Consider

$$\int [dq] q(t_1) \dots q(t_N) e^{iS} \quad (6.18)$$

The N functions do not need to be put in a specific order, as they commute. However, when we repeat the steps above, we have to time order them to be able to split the integrals correctly. This means that

$$\int [dq] q(t_1) \dots q(t_N) e^{iS} = \langle q_i T_i | T \{ q(t_1) \dots q(t_N) \} | q_f T_f \rangle \quad (6.19)$$

6.2 From QM to QFT

the generalization, at least to a formal level, is quite straightforward: one writes

$$\langle \phi_f T_f | \phi_i T_i \rangle = \int_{\phi(x, T_i) = \phi_i(x)}^{\phi(x, T_f) = \phi_f(x)} \mathcal{D}\phi e^{i \int d^4x \mathcal{L}} \quad (6.20)$$